Neurocomputing 700 (2026) 134375

Contents lists available at ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom

MPGTimer: A long-context multi-scale patch and graph-enhanced transformer for time series forecasting

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HIGHLIGHTS

- MPGTimer, a Transformer-based model for long-context time series forecasting.
- Multi-scale Patch Integration builds hierarchical temporal representations.
- Graph-enhanced Interaction Attention models cross-variable dependencies.
- Achieves 3.13% average improvement over state-of-the-art methods.

ARTICLE INFO

Dataset link: Time-Series-Library dataset (Reference data)

Communicated by F.G. Guimaraes

Keywords:
Long-context time series forecasting
Transformer
Multi-scale patch integrator
Graph-enhanced interaction attention

ABSTRACT

Time series forecasting aims to model temporal dynamics from historical observations in order to predict future trends. As the scale of real-world data continues to grow, effectively exploiting long contextual information has become increasingly important. However, long contexts often exhibit hierarchical temporal structures and time-varying, non-uniform dependencies across variables, which are difficult to model within a unified framework. As a result, the performance of existing approaches is often constrained. To address these challenges, we propose MPGTimer, a multivariate long-sequence modeling framework tailored for long-context forecasting. MPGTimer introduces a Multi-scale Patch Integrator (MPI) to extract local segments at multiple temporal resolutions and align them into a unified representation. Building upon this design, the model incorporates a Graph-enhanced Interaction Attention (GIA) mechanism to capture both intra- and inter-variable dependencies over flattened patches. Positional embeddings are further employed to reinforce temporal causality and variable equivalence. In addition, a graph-structured interaction layer is used to adaptively learn correlation weights among variable nodes, allowing for more effective modeling of complex variable relationships. Extensive experiments on multiple real-world datasets demonstrate that MPGTimer achieves an average 3.13% improvement in forecasting accuracy over state-of-the-art methods.

1. Introduction

Multivariate time series forecasting (MTSF) focuses on modeling historical observations from multiple variables to predict their future trends. It has become a fundamental component of modern data analysis and plays a critical role in a wide range of applications, including commercial [5,9,28] and industrial [4] domains. With the continuous growth of data volume, time series increasingly exhibit long-range dependencies, making long-context modeling an important research direction. However, prior studies have shown that simply extending the lookback length of Transformer-based models does not lead to consistent improvements in forecasting performance [23,39]. One key reason

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https://doi.org/10.1016/j.neucom.2026.134375
Received 27 March 2026; Received in revised form 7 June 2026; Accepted 27 June 2026
Available online 4 July 2026
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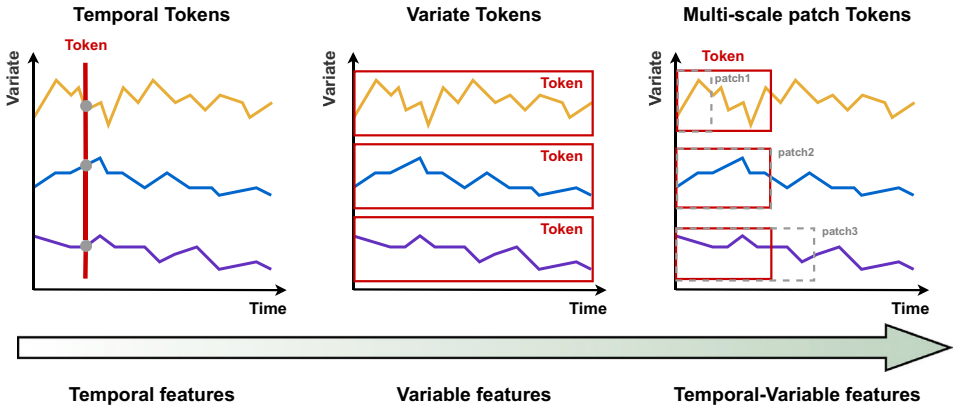


Fig. 1. Tokenization strategies for long-context time series modeling. Time-step-based tokens (left), variable-based tokens (center), and the proposed multi-scale patch-based tokens that balance both perspectives (right).

is that as the number of observations increases, dependency structures within the sequence become more complex, while the scalability of existing architectures for long-context modeling remains limited. Moreover, most existing approaches struggle to jointly capture multi-scale temporal patterns and complex interactions across variables, even though such interactions play a crucial role in determining forecasting accuracy under long-context settings [32,36].

Early deep learning approaches, such as recurrent neural networks (RNNs) [38] including LSTM [7] and GRU [3], introduced memory mechanisms to improve sequence modeling capability. However, they still suffer from vanishing gradients and inefficient training over long temporal horizons. More recently, the Transformer architecture [29] has been widely adopted for time series forecasting, with representative models such as Informer [41] and Autoformer [34]. These methods typically treat each time step as an individual token, which enables effective modeling of short-term dependencies but makes it difficult to stably capture long-range temporal trends. To better model long-term dependencies along the variable dimension, iTransformer [17] proposes a dimension-inverted representation, shifting from time-step tokens to variable-level tokens. While this design improves the modeling of long-term temporal dependencies, it inevitably weakens the representation of local temporal patterns and limits the modeling of both intra- and inter-variable interactions.

As illustrated in Fig. 1, to address the above challenges and improve the modeling of multivariate time series under long-context settings, we propose MPGTimer. The model leverages multi-scale patch embeddings to capture short-term local patterns and long-term trend information. In addition, graph-enhanced patch interaction attention is introduced to model dynamic dependencies among variables, thereby improving forecasting performance and generalization for long multivariate sequences.

The contributions of this paper are as follows:

- We propose MPGTimer, a Transformer-based framework tailored for long-context time series forecasting. It enhances long-context representation by jointly constructing multi-scale input representations and modeling time-varying dependencies.
- We introduce a Multi-scale Patch Integrator (MPI) and a Graph-enhanced Interaction Attention (GIA) mechanism. MPI constructs multi-scale patch embeddings, while GIA models patch interactions via structured masks and bias terms, and captures multivariate dependencies through graph-enhanced attention with learnable soft weights.
- Extensive experiments on seven publicly available datasets demonstrate that MPGTimer establishes a new state-of-the-art benchmark in long-context forecasting scenarios, achieving an average accuracy improvement of 3.13%.

2. Related work

Time series forecasting has experienced rapid progress in recent years, largely driven by advances in deep learning. In the early stages, when computational resources were limited, forecasting methods mainly relied on traditional statistical models. Representative examples include the Autoregressive Moving Average (ARMA) family [2], which typically performs well in short-term forecasting but exhibits limited modeling capacity under complex settings such as non-stationarity and multivariate scenarios. To address these limitations, deep learning architectures have been increasingly incorporated into time series forecasting. Recurrent neural networks (RNNs) [38] and their variants, including LSTM [7], GRU [3], bidirectional recurrent neural networks (BiRNN) [27], and deep recurrent neural networks [6], significantly improve the modeling of temporal dependencies. Building on this foundation, subsequent studies have further combined deep learning with classical statistical modeling principles to develop more expressive forecasting models. For example, DeepAR [26] integrates the probabilistic formulation of traditional autoregressive (AR) models with RNNs, enabling more effective modeling of nonlinear dynamics. LSTNet [11] combines one-dimensional convolutional networks with GRU modules to better capture local and periodic patterns. TCN [1] eliminates recurrent structures by adopting causal and dilated convolutions, which expand the receptive field and enhance training stability. Despite these advances, most existing approaches rely on a single temporal perspective. Their ability to capture multi-scale features and complex dynamic patterns remains limited, which constrains overall model expressiveness.

Beyond deep neural network-based approaches, simple and efficient linear models have also attracted increasing attention. DLinear [39] decomposes a time series into trend and residual components and applies linear layers for direct multi-step forecasting. Following this line of work, RLinear [13] introduces Reversible Instance Normalization (RevIN) and a channel-independent (CI) design within a single linear mapping, improving the modeling of multi-period components in long-term time series. In addition, convolutional neural networks (CNNs) have been employed for time series modeling. TimesNet [33] combines one-dimensional convolutions with frequency-domain filters to extract temporal patterns at multiple granularities, thereby improving multi-scale representation learning. SCNet [35] integrates spectral energy prioritization with multi-scale convolutions to effectively improve multivariate time series classification performance. Studies that integrate the above two categories of methods have also begun to emerge. STNet [12] employs EWMA-based decomposition and combines multi-scale convolution with linear modules to effectively model both trend and seasonal components. Mamba-based state space model architectures have also gradually been studied. TimePro [22] constructs variate- and time-aware hyper-states to efficiently model long-term temporal

Table 1
Summary of representative multivariate time series forecasting models in terms of architecture, tokenization strategy, and temporal and variable dependency modeling. Differences in temporal and variable modeling strengths reflect their ability to capture complex temporal patterns and variable relationships. The graph column indicates whether graph structures are incorporated for dependency modeling, while the star ratings provide an overall assessment of forecasting performance.

Model	Architecture	Tokenization	Temporal	Variable	Graph	Overall performance
Autoformer(2021)	Transformer	Sequence	High	Low	✗	★★☆☆☆
TimesNet(2023)	CNN	Sequence	High	Medium	✗	★★★☆☆
DLinear(2023)	Linear	Sequence	Medium	Low	✗	★★☆☆☆
Crossformer(2023)	Transformer	Segment	Medium	High	✗	★★★☆☆
iTransformer(2024)	Transformer	Variable	High	Medium	✗	★★★☆☆
Timer(2024)	Transformer	Sequence	High	Medium	✗	★★★☆☆
GAPatch(2025)	CNN	Single-scale patch	Medium	High	✓	★★★★☆
Timer-XL(2025)	Transformer	Single-scale patch	High	High	✗	★★★★☆
MPGTimer(Ours)	Transformer	Multi-scale patch	High	High	✓	★★★★★

dependencies, achieving competitive long-term forecasting performance while maintaining linear complexity.

In recent years, Transformer-based methods have brought substantial progress to time series forecasting, largely owing to the self-attention mechanism. Informer [41] alleviates the computational bottlenecks of vanilla Transformers in both short- and long-term forecasting by introducing probabilistic attention, attention distillation, and a generative decoder; however, structural temporal patterns are not explicitly modeled. In response to this limitation, Autoformer [34] introduces a trend-seasonality decomposition architecture and an autocorrelation mechanism to explicitly capture long-term trends and periodic fluctuations. Stationary [19] explicitly addresses non-stationarity, improving stability in long-horizon forecasting. Crossformer [40] introduces dimension-segment-wise embedding, two-stage attention, and a hierarchical encoder–decoder architecture to enhance multivariate time series modeling, but it decouples temporal and variable dependencies, limiting unified dependency modeling. iTransformer [17] treats the entire time series of each variable as a token via dimension inversion, improving long-term dependency modeling but lacking explicit inter-variable dependency modeling. TQNet [15] employs periodic learnable queries (TQ) to capture global variable dependencies, while integrating single-layer attention and lightweight MLPs for efficient forecasting. UniTST [16] explores a unified token representation across different forecasting tasks to improve generalization. More recently, large-scale pre-training frameworks have been introduced to time series modeling. DDformer [14] models complex inter-variable relationships and temporal variations through dynamic variable fusion and a dynamic difference attention mechanism. GAPatch [8] models local temporal dynamics with overlapping patches and captures spatial dependencies via static and dynamic graphs using graph attention networks. EIAformer [31] employs hybrid decomposition and dilated causal convolution to capture multiscale information, and introduces a channel fusion mechanism to model inter-variable dependencies. FreEformer[37] leverages global frequency-domain properties and an enhanced attention mechanism with an additional learnable matrix and L1 normalization to achieve more robust multivariate time series forecasting. CSformer[30] adopts a strategy of channel independence followed by mixing through a two-stage multiheaded self-attention mechanism, integrated with adapter techniques to achieve robust multivariate time series forecasting. ReCast [21] quantizes local patterns through a codebook-driven patch mechanism and achieves lightweight time series forecasting via a residual dual-branch structure and a reliability-aware update mechanism. In long-context forecasting, Timer [20] pioneers a decoder-only architecture for time series pre-training, enabling autoregressive modeling with flexible input and output lengths across variables. Building upon this framework, Timer-XL [18] incorporates masking mechanisms to extend the context window; however, its ability to jointly capture short- and long-term temporal patterns remains limited. To more clearly illustrate the design differences among representative methods, we summarize and compare their key modeling characteristics in Table 1.

Motivated by these observations, we propose MPGTimer, which improves long-context forecasting by combining multi-scale input representations with time-varying dependency modeling.

3. MPGTimer

Problem Definition. The task of MPGTimer is long-context multivariate time series forecasting. Unlike the standard forecasting settings commonly adopted in prior work, MPGTimer focuses on prediction under substantially longer contexts. Compared with conventional forecasting tasks, long-context multivariate forecasting is considerably more challenging. Specifically, a model must simultaneously capture short-term dynamics within local temporal windows and extract structural temporal information from long-span historical sequences. Moreover, as the context length increases, the statistical properties of the series may evolve over time, further exacerbating the modeling difficulty. The core challenge of this task lies in efficiently modeling temporal dependencies over long horizons, stably characterizing variable relationships, and balancing predictive accuracy with computational efficiency and robustness. Formally, let the historical multivariate time series be denoted as in Eq. (1), where C is the number of variables and L represents the context length. The forecasting target is defined as in Eq. (2), corresponding to the future sequence over the next H time steps.

$$\mathbf{X} = \{\mathbf{x}_0, \dots, \mathbf{x}_{L-1}\} \in \mathbb{R}^{C\times L} \tag{1}$$

$$\mathbf{Y} = \{\mathbf{x}_L, \dots, \mathbf{x}_{L+H-1}\} \in \mathbb{R}^{C\times H} \tag{2}$$

A key difficulty under long-context settings is enabling the model to suppress irrelevant historical noise while preserving informative long-term dependencies, so as to accurately capture variable interactions and temporal patterns across multiple time scales in real-world scenarios.

Structure Overview. MPGTimer is a novel framework for long-context multivariate time series forecasting. The overall architecture of MPGTimer is illustrated in Fig. 2. Unlike existing Transformer-based methods that rely on fixed-scale patch partitioning, MPGTimer introduces a Multi-scale Patch Integrator (MPI) at the input stage and incorporates Graph-enhanced Interaction Attention (GIA), enabling the efficient modeling of dependencies across both the temporal and variable dimensions.

During sequence construction, MPGTimer introduces a multi-scale patch embedding scheme. The raw input sequence is segmented into patches of different lengths, each representing a contiguous temporal interval at a specific scale. These patches are generated in parallel and projected into a shared latent space, where token counts are aligned via upsampling or downsampling. The resulting multi-scale representations preserve short-term fluctuations and long-term temporal evolution

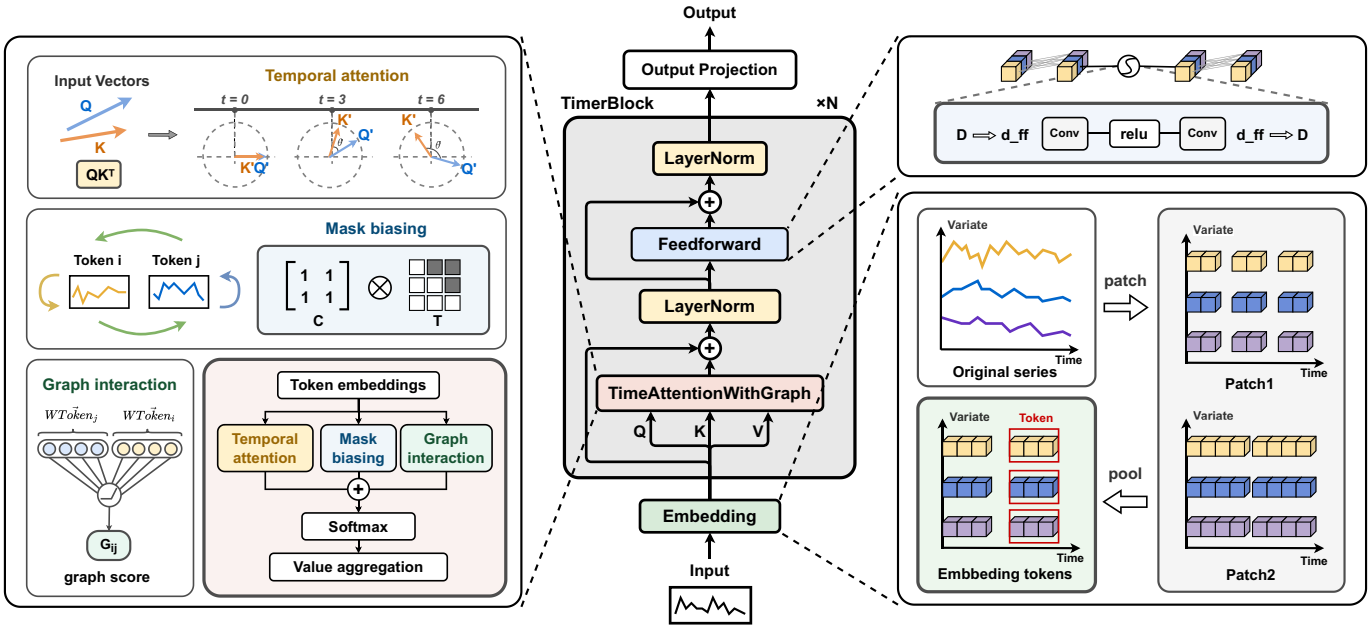


Fig. 2. An overview of the MPGTimer architecture. The input sequence is first partitioned into patches of multiple scales using the multi-scale patch integrator (MPI), with token numbers aligned through sampling. A decoder-only, multi-layer TimerBlock is then employed to construct sparse temporal attention, enabling global modeling of long-range dependencies. In addition, graph-enhanced interaction attention (GIA) is integrated to capture dynamic variable relationships, further improving forecasting performance under long-context settings.

within a unified embedding space, providing a solid basis for subsequent dependency modeling. For joint temporal-variable modeling, MPGTimer adopts a Decoder-only architecture and builds long-sequence attention modules based on stacked TimerBlocks proposed in prior work [18]. Within each TimerBlock, multivariate long sequences are jointly modeled along both the temporal and variable dimensions. Along the temporal dimension, multi-scale patch tokens are processed under causal constraints to capture time-dependent relationships. A time-priority token ordering strategy, together with variable dependency graphs and temporal causal masks, is used to construct a unified attention mask. This design enforces strict temporal causality while enabling coordinated modeling of variable interactions. Along the variable dimension, MPGTimer further introduces a dual-path content-based relation modeling mechanism. This mechanism combines a fully connected projection path with a graph-enhanced attention path to adaptively characterize variable correlations. The fully connected path aggregates global patterns across temporal segments, while the graph-enhanced relation modeling path assigns continuous soft weights at the token level to capture dynamic variable dependencies. Outputs from both paths are fused via residual connections and normalization, and are subsequently mapped back to the time-series representation space. Therefore, given the multivariate time series $\mathbf{X} \in \mathbb{R}^{C\times L}$, our goal is to predict the future sequence $\mathbf{Y} \in \mathbb{R}^{C\times H}$. The overall model architecture can be formulated as follows:

$$
\begin{aligned}
&\mathbf{X}^{(s)} = \text{Patchify}_{p_s}(\mathbf{X}), \quad s = 1, \dots, S \\
&\mathbf{Z} = \text{Fuse}\left(\{\text{Align}(\text{Embed}(\mathbf{X}^{(s)}))\}_{s=1}^{S}\right) \\
&\mathbf{H}^{i} = \text{TimerBlock}_i(\mathbf{H}^{i-1}), \quad i = 1, \dots, D \\
&\mathbf{R} = \text{Norm}\left(\mathbf{H} + \text{FC}(\mathbf{H}) + \text{GraphAttn}(\mathbf{H})\right) \\
&\hat{\mathbf{Y}} = \text{Head}(\mathbf{H}^{D})
\end{aligned}
\tag{3}
$$

In the above formulation, $\text{Patchify}_{p_s}(\cdot)$ denotes the segmentation operation of the multi-scale Patch Integrator (MPI) at the s-th scale, while $\text{Fuse}(\cdot)$ represents the fusion of multi-scale information to obtain a unified representation $\mathbf{Z}$. The fused representation is then fed into an encoder composed of stacked TimerBlock layers for modeling. For variable modeling, the graph-enhanced interaction attention (GIA) consists of two components, where FC captures content-based global dependencies, and GraphAttn denotes an attention mechanism that incorporates variable dependencies and temporal causality. The two paths are fused via residual connections and normalization, denoted by $\text{Norm}(\cdot)$. Finally, the prediction head $\text{Head}(\cdot)$ maps the high-level representation to the target sequence $\hat{\mathbf{Y}}$.

Multi-scale Patch Integrator. In long-context time series forecasting, input sequences typically contain both short-term variations and long-term trends. Most Transformer-based methods adopt fixed-length patch segmentation, where a time series of length L is uniformly divided into L/P patch tokens using a window size of P. Although this strategy reduces the computational burden of long-sequence modeling, operating at a single temporal scale limits the model's ability to capture multi-scale dependencies. As a result, fine-grained local patterns may be overlooked, while long-range global trends are difficult to align. To overcome this limitation, MPGTimer introduces a Multi-scale Patch Integrator (MPI) at the input stage. Unlike existing methods that mainly model multi-scale information implicitly through unified token representations or internal model architectures, MPI directly constructs multi-scale representations at the input level using patches of different lengths. This design enables the model to more explicitly integrate short-term and long-term temporal information. Formally, given an input sequence $\mathbf{X} = (\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{L-1}) \in \mathbb{R}^{C\times L}$, where C denotes the number of variables and L denotes the sequence length, MPI then introduces a set of patch scales $\{k_1, k_2, \dots, k_M\}$ and partitions the historical sequence independently at each scale:

$$
\mathbf{X}^{(m)} = \left(\mathbf{X}_1^{(m)}, \mathbf{X}_2^{(m)}, \dots, \mathbf{X}_{N_m}^{(m)}\right), \quad \mathbf{X}_i^{(m)} \in \mathbb{R}^{C\times k_m} \tag{4}
$$

Here, k_m denotes the patch length at the m-th scale, and $N_m = \lfloor L/k_m \rfloor$ is the corresponding number of tokens. Accordingly, smaller values of k_m produce patches that capture local temporal variations, whereas larger k_m retain trend information over longer time spans.

After multi-scale patches are obtained, MPI projects each patch into a shared embedding space via a linear transformation. Since the number

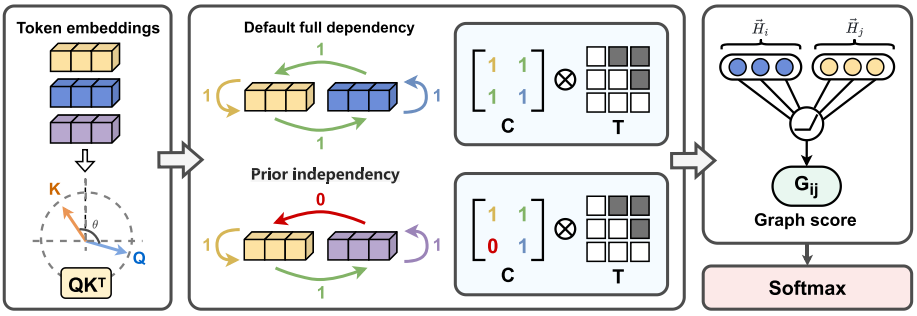


Fig. 3. TimeAttentionWithGraph. An attention module that integrates rotary positional encoding, a joint structural mask, and graph-enhanced attention to capture temporal dependencies and variable structural relations in long-context sequences at the patch level.

of tokens N_m differs across scales, a scale-alignment operation is applied to map all embeddings to a common token length N. Specifically, when $N_m > N$, the temporal dimension is downsampled by average pooling; when $N_m < N$, linear interpolation is used for upsampling. Since N is indirectly determined by the patch size, we further analyze different values of N by varying the patch size in the hyperparameter study to evaluate the optimal choice of N under different forecasting tasks. It should be noted that the above alignment operations are performed on patch-level tokens, where each token has already encoded the local semantic information within its corresponding temporal window. Therefore, average pooling or linear interpolation along the token dimension can be viewed as a semantic-space resampling process, enabling scale unification while preserving the consistency of the overall temporal structure. Finally, the aligned multi-scale embeddings are concatenated along the feature dimension and fused through a linear projection, forming a unified multi-scale representation that serves as the input to subsequent temporal attention modules. By integrating multi-scale temporal patterns at the input stage, MPI enables the model to jointly capture short-term fluctuations and long-range dependencies, providing richer and more robust representations for downstream attention modeling.

Graph-enhanced Interaction Attention. In long-context settings, the growing number of tokens makes approaches that rely solely on temporal modeling less effective at preserving structural relationships among variables. To improve representation quality, it is therefore essential to jointly model long-range temporal dependencies while maintaining awareness of variable structures. Motivated by this observation, MPGTimer extends the multi-layer TimerBlock proposed in prior work by introducing the TimeAttentionWithGraph module. By incorporating variable-structure biases and graph-enhanced attention, this module jointly models temporal dependencies and variable interactions within the patch-level token space. The overall architecture of the module is illustrated in Fig. 3.

The modeling of temporal dependencies is centered on the content-based attention term $\mathbf{QK}^\top$. Unlike approaches that inject positional information by directly adding positional embeddings to the input features, MPGTimer incorporates rotary positional embeddings during the construction of the query and key vectors. This design encodes temporal position information into the orientation of the $\mathbf{Q}$ and $\mathbf{K}$ vectors rather than their magnitudes. Specifically, the patch-level token representations $\mathbf{H}$ are linearly projected to obtain $\mathbf{Q}$ and $\mathbf{K}$, which are then rotated according to their temporal position indices. As a result, tokens with identical content but occurring at different time steps are represented by directionally distinct vectors. The resulting inner product $\mathbf{QK}^\top$ therefore naturally captures relative temporal relationships. Building on this formulation, the TimeAttentionWithGraph module further models cross-patch dependencies under causal constraints by introducing structured attention masks. These constraints enable the joint modeling of temporal dependencies and variable-level reachability within a unified attention framework. Patch tokens are first flattened into a one-dimensional sequence in time-priority order, ensuring that attention computation strictly respects temporal causality and prevents information leakage from future time steps.

To systematically model the attention masks in multivariate time series, MPGTimer introduces a joint structure that considers both the variable and temporal dimensions. In the variable dimension, a variable dependency graph $\mathbf{A} \in \mathbb{R}^{C\times C}$ is used to describe the structural dependency among variables, where C denotes the number of variables. In the temporal dimension, a causal mask $\mathbf{T} \in \mathbb{R}^{N\times N}$ constrains each token to attend only to the current and preceding time steps, where N denotes the number of tokens corresponding to each variable. In the absence of additional structural priors, all variable pairs are allowed to attend to each other, and $\mathbf{A}$ is defined as a fully connected matrix with $A_{m,n} = 1$. When explicit prior constraints are available, the corresponding entries are set to $A_{m,n} = 0$. The final attention mask is then constructed via the Kronecker product ($\otimes$):

$$\mathbf{M} = \mathbf{A} \otimes \mathbf{T} \in \mathbb{R}^{(CN)\times(CN)} \tag{5}$$

This mask specifies the spatio-temporal attention scope of each token.

Since the joint attention mask $\mathbf{M}$ imposes hard constraints through a binary structural design, TimeAttentionWithGraph further introduces a graph-enhanced term $\mathbf{G}$ to capture fine-grained variations in dependency strength by assigning continuous soft weights to token interactions. This results in a complementary mechanism that combines structural constraints with similarity-based modeling. Unlike conventional graph attention networks (GATs), which typically learn trainable adjacency matrices or edge weights, the proposed method does not introduce independent graph structure parameters. Instead, token representations are projected through shared linear transformations, and pairwise similarities are computed in the projected feature space to characterize dynamic dependencies. Specifically, patch-level token representations in temporal attention are treated as graph nodes, where each token corresponds to a variable–temporal patch pair. The node features are projected into a graph-aware attention space through linear transformations, and the similarity scores between arbitrary node pairs are subsequently computed. Accordingly, the graph-enhanced term $\mathbf{G} \in \mathbb{R}^{(CN)\times(CN)}$ is formulated as follows:

$$G_{i,j} = \tilde{\mathbf{H}}_i^\top \tilde{\mathbf{H}}_j, \quad \tilde{\mathbf{H}} = \mathbf{H}\mathbf{W}_g \tag{6}$$

The resulting score reflects the similarity between different tokens and can be interpreted as a soft characterization of the dynamic dependency strength among tokens. Finally, the content-driven temporal attention term $\mathbf{QK}^\top$, the joint structural mask $\mathbf{M}$, and the graph-enhanced attention term $\mathbf{G}$ are integrated within a unified attention space. The resulting attention formulation is given below:

$$\mathbf{S} = \mathbf{QK}^\top + \mathbf{M} + \mathbf{G} \tag{7}$$

$$\text{TimeAttentionWithGraph}(\mathbf{H}) = \text{Softmax}\left(\frac{\mathbf{S}}{\sqrt{d_k}}\right)\mathbf{H}\mathbf{W}^v \tag{8}$$

The above fusion can be viewed as an additive decomposition in the attention logits space, where $\mathbf{QK}^\top$ represents the content-driven similarity term, $\mathbf{M}$ serves as a structural constraint encoding token reachability, and $\mathbf{G}$ continuously models dependency strength within the reachable regions. This additive formulation is consistent with the bias injection mechanism commonly adopted in standard attention architectures, allowing information from different sources to contribute to the final attention distribution in an independent and interpretable manner. Consequently, the proposed unified modeling strategy enables temporal dynamics, structural priors, and inter-variable interaction strengths to be jointly represented within a shared attention space, thereby enhancing representation capability in long-context forecasting scenarios.

4. Experiments

To thoroughly evaluate the forecasting performance of MPGTimer, we conduct extensive experiments on a range of benchmark datasets. Unlike conventional long-term forecasting settings, our evaluation focuses on long-context forecasting. Specifically, we fix the prediction horizon while extending the lookback length to weekly- and monthly-scale contexts.

Datasets. To evaluate MPGTimer across a broad range of real-world scenarios, we conduct experiments on seven publicly available multivariate time series datasets. These datasets include four ETT benchmarks (ETTm1, ETTm2, ETTh1, and ETTh2), as well as the Electricity (ECL) and Weather datasets utilized in Informer [41], and the Solar-Energy dataset utilized in LSTNet [11]. These datasets cover diverse application domains and temporal resolutions, and exhibit rich periodicity, long-term trends, and pronounced non-stationary characteristics. To ensure a fair and consistent comparison, we strictly follow the standard training, validation, and test sets adopted by Timer_XL [18] for all datasets, and evaluate all models under the same experimental protocol. Detailed descriptions of each dataset are provided in Appendix A.1.

Baselines. We consider two groups of representative forecasting models under different experimental settings. For the long-context setting which is the primary focus of our work, we select eight widely recognized time series forecasting models as benchmarks, including six Transformer-based models: Timer_XL [18], Timer [20], UniTST [16], iTransformer [17], Stationary [19], and Autoformer [34]; the CNN-based TimesNet [33]; and the linear projection-based DLinear [39]. Under the standard short-context setting, several recent state-of-the-art methods, including ReCast [21], FreEformer [37], TQNet [15], CSformer [30], TimePro and iTransformer [17], are further introduced to maintain consistency with commonly adopted benchmarks and to validate the generalization capability of the proposed model. Detailed baseline configurations are provided in Appendix A.2. To ensure fair comparison and reproducibility, all methods are evaluated under consistent data splits and evaluation protocols. For several approaches that are particularly sensitive to training strategies and hyperparameter settings, performance may vary across different experimental environments. Therefore, the results of baseline methods are primarily reported using the values provided in their original papers to maintain consistency with prior studies. In addition, representative methods are re-implemented under a unified experimental setting in supplementary analyses such as hyperparameter studies, where all models are evaluated under the same configuration as MPGTimer to enable a more controlled and equitable comparison. Our model is evaluated against these baselines under identical experimental settings. All experiments are implemented using Python 3.8.8 [25] and PyTorch 2.0.1 (CUDA 11.7) [24], and executed on a 40-core Intel Xeon Silver 4210R CPU paired with a single NVIDIA GeForce RTX 3090 GPU. The training process employs the Adam optimizer [10] and incorporates an Early Stopping strategy, where training is automatically halted if the validation loss does not improve for three consecutive epochs, preventing overfitting. Detailed hyperparameter settings are listed in Appendix A.3. Consistent with prior work, we adopt Mean Squared Error (MSE) and Mean Absolute Error (MAE) as evaluation metrics.

4.1. Long-context forecasting results

MPGTimer achieves superior performance on long-context multivariate time series forecasting tasks. These improvements are primarily observed under long-context settings with extended lookback lengths. The experimental results are summarized in Table 2, where the best results are highlighted in bold red and the second-best results are underlined in blue. Notably, across the 35 reported MSE settings, MPGTimer ranks among the top two in 34 cases and attains the best performance in 27 cases. These results establish a new benchmark in long-context forecasting accuracy.

Relative to the state-of-the-art long-context forecasting model Timer_XL, MPGTimer achieves reductions of 3.13% in MSE and 1.74% in MAE. When compared with the Transformer-based iTransformer, the reductions increase to 9.30% and 7.55%. For the linear projection-based DLinear, MPGTimer achieves reductions of 12.99% and 9.51%, and for the CNN-based TimesNet, the reductions further reach 16.43% and 11.15%. In addition, MPGTimer exhibits particularly strong performance on the Weather and ETTm2 datasets when compared with Timer_XL. On the Weather dataset, MSE and MAE are reduced by 5.42% and 2.93%, respectively, while on ETTm2 the reductions reach 5.17% and 1.86%.

Specifically, except for the Electricity dataset, MPGTimer achieves predictive performance that is comparable to or better than current state-of-the-art long-context models across the remaining six datasets. In the Electricity dataset, most variables exhibit highly similar periodic patterns along the temporal dimension, and their correlations are largely driven by exogenous time-related factors, such as daily cycles and weekday effects. As a result, the observed dependencies mainly reflect temporal alignment rather than cross-variable structural interactions. MPGTimer is explicitly designed to capture dynamic cross-variable structures through graph-enhanced attention. However, in the Electricity dataset, which is primarily time-dominant, the additional modeling of inter-variable interactions is less beneficial. Instead, it increases the complexity of the attention space, which may slightly hinder forecasting performance. In contrast, Timer-XL places greater emphasis on temporal modeling, allowing it to aggregate time-dependent information more directly and thus achieve more stable forecasting behavior on this dataset. This observation suggests that MPGTimer is particularly well-suited to scenarios with pronounced dynamic variable interactions and time-varying dependency structures. In contrast, its advantages may be less pronounced in settings dominated by synchronous periodic patterns with weak variable dependencies.

These experimental results indicate that MPGTimer offers practical value across the majority of real-world datasets. The observed performance gains are primarily attributed to its architectural design. Firstly, the Multi-scale Patch Integrator extracts patch-level representations across different temporal resolutions, enabling the model to jointly capture short-term fluctuations and long-term trends. In contrast, iTransformer treats the entire time series of each variable as a single token for attention modeling, without patch-level reorganization. In long-context settings, this design choice in iTransformer leads to rapidly growing token counts and consequently higher computational and modeling overhead. Moreover, the Graph-enhanced Interaction Attention assigns continuous soft weights to cross-variable dependencies by computing similarity scores between patch-level token representations. By comparison, Timer_XL focuses primarily on long-range temporal modeling, while variable structural constraints are imposed only through binary structural masks. Without soft weighting, variable interaction strengths cannot adapt to time and context, limiting the modeling of complex dependencies. These architectural differences highlight the stronger representational capacity of MPGTimer in jointly modeling multi-scale

Table 2
Benchmark experiments for long-term multivariate forecasting (672-pred-{96,192,336,720}). Under a single model, experiments were conducted for four forecast horizons using rolling forecasts.

Models		MPGTimer (Ours)		Timer_XL (2025)		Timer (2024)		UniTST (2024)		iTransformer (2024)		DLinear (2023)		TimesNet (2023)		Stationary (2022)		Autoformer (2021)	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MAE	MAE
ETTh1	96	**0.355**	**0.389**	0.364	0.397	0.371	0.404	0.379	0.415	0.387	0.418	0.369	0.400	0.452	0.463	0.452	0.478	0.467	0.499
	192	**0.390**	**0.412**	0.405	0.424	0.407	0.429	0.415	0.438	0.416	0.437	0.405	0.422	0.474	0.477	0.484	0.510	0.492	0.523
	336	**0.406**	**0.424**	0.427	0.439	0.434	0.445	0.440	0.454	0.434	0.450	0.435	0.445	0.493	0.489	0.511	0.522	0.519	0.531
	720	**0.419**	**0.447**	0.439	0.459	0.461	0.466	0.482	0.482	0.447	0.473	0.493	0.508	0.560	0.534	0.571	0.543	0.589	0.560
	Avg	**0.393**	**0.418**	0.409	0.430	0.418	0.436	0.429	0.447	0.421	0.445	0.426	0.444	0.495	0.491	0.505	0.513	0.517	0.528
ETTh2	96	**0.274**	**0.341**	0.277	0.343	0.285	0.344	0.343	0.398	0.304	0.362	0.305	0.371	0.340	0.374	0.348	0.403	0.358	0.397
	192	**0.333**	**0.382**	0.348	0.391	0.365	0.400	0.376	0.420	0.372	0.407	0.412	0.439	0.402	0.414	0.408	0.448	0.435	0.451
	336	**0.360**	**0.406**	0.375	0.418	0.412	0.440	0.399	0.435	0.418	0.440	0.527	0.508	0.452	0.452	0.424	0.457	0.454	0.475
	720	**0.401**	**0.440**	0.409	0.458	0.468	0.487	0.419	0.457	0.463	0.476	0.830	0.653	0.462	0.468	0.448	0.476	0.479	0.492
	Avg	**0.342**	**0.392**	0.352	0.402	0.382	0.418	0.384	0.428	0.389	0.421	0.518	0.493	0.414	0.427	0.407	0.446	0.431	0.454
ETTm1	96	**0.280**	0.339	0.290	0.341	0.281	**0.338**	0.289	0.348	0.311	0.365	0.307	0.350	0.338	0.375	0.414	0.414	0.466	0.466
	192	**0.327**	**0.368**	0.337	0.369	0.330	**0.368**	0.332	0.375	0.353	0.390	0.337	**0.368**	0.371	0.387	0.524	0.482	0.504	0.496
	336	**0.364**	0.390	0.374	0.392	0.367	0.393	0.365	0.397	0.387	0.411	0.366	**0.387**	0.410	0.411	0.541	0.497	0.574	0.530
	720	0.424	0.425	0.437	0.428	0.432	0.433	0.421	0.431	0.452	0.445	**0.419**	**0.419**	0.478	0.450	0.578	0.509	0.596	0.558
	Avg	**0.349**	**0.381**	0.359	0.382	0.352	0.383	0.352	0.388	0.376	0.403	0.357	**0.381**	0.399	0.406	0.514	0.475	0.535	0.512
ETTm2	96	0.169	**0.256**	0.175	0.257	0.175	0.257	0.171	0.260	0.183	0.272	**0.167**	0.263	0.187	0.267	0.237	0.306	0.255	0.339
	192	**0.224**	0.295	0.242	0.301	0.239	0.301	0.228	**0.230**	0.250	0.315	0.230	0.311	0.249	0.309	0.330	0.387	0.279	0.335
	336	**0.274**	**0.329**	0.293	0.337	0.293	0.342	0.282	0.336	0.311	0.356	0.298	0.361	0.321	0.351	0.404	0.424	0.331	0.374
	720	**0.361**	**0.385**	0.376	0.390	0.392	0.407	0.380	0.398	0.417	0.419	0.432	0.446	0.497	0.403	0.525	0.486	0.413	0.450
	Avg	**0.257**	0.316	0.271	0.322	0.275	0.327	0.265	**0.306**	0.290	0.340	0.282	0.345	0.314	0.333	0.374	0.401	0.320	0.374
Electricity	96	**0.125**	0.220	0.127	**0.219**	0.129	0.221	0.130	0.225	0.133	0.229	0.138	0.238	0.184	0.288	0.185	0.287	0.256	0.357
	192	**0.144**	0.238	0.145	**0.236**	0.148	0.239	0.150	0.244	0.158	0.258	0.152	0.251	0.192	0.295	0.282	0.368	0.291	0.376
	336	0.160	0.254	**0.159**	**0.252**	0.164	0.256	0.166	0.262	0.168	0.262	0.167	0.268	0.200	0.303	0.289	0.377	0.290	0.379
	720	0.196	0.286	**0.187**	**0.277**	0.201	0.289	0.206	0.297	0.205	0.294	0.203	0.302	0.228	0.325	0.305	0.399	0.320	0.403
	Avg	0.156	0.250	**0.155**	**0.246**	0.161	0.251	0.163	0.257	0.164	0.258	0.165	0.265	0.201	0.303	0.265	0.358	0.289	0.379
Weather	96	**0.150**	**0.200**	0.157	0.205	0.151	0.202	0.152	0.206	0.174	0.225	0.169	0.229	0.169	0.228	0.185	0.241	0.355	0.409
	192	**0.196**	**0.243**	0.206	0.250	**0.196**	0.245	0.249	0.198	0.227	0.268	0.211	0.268	0.222	0.269	0.286	0.325	0.421	0.450
	336	**0.247**	**0.283**	0.259	0.291	0.249	0.288	0.251	0.291	0.290	0.309	0.258	0.306	0.290	0.310	0.323	0.347	0.452	0.465
	720	**0.316**	**0.332**	0.337	0.344	0.330	0.344	0.322	0.340	0.374	0.360	0.320	0.362	0.376	0.364	0.436	0.401	0.513	0.496
	Avg	**0.227**	**0.265**	0.240	0.273	0.232	0.270	0.231	0.272	0.266	0.291	0.239	0.291	0.264	0.293	0.308	0.329	0.435	0.455
Solar-Energy	96	0.163	**0.213**	**0.162**	0.221	0.212	0.230	0.190	0.240	0.183	0.265	0.193	0.258	0.180	0.272	0.199	0.290	0.206	0.296
	192	0.189	**0.233**	**0.187**	0.239	0.232	0.246	0.223	0.264	0.205	0.283	0.214	0.274	0.199	0.286	0.243	0.307	0.254	0.328
	336	0.206	**0.249**	**0.205**	0.255	0.237	0.253	0.250	0.283	0.224	0.299	0.233	0.291	0.220	0.301	0.264	0.322	0.272	0.330
	720	**0.235**	0.274	0.238	0.279	0.252	**0.266**	0.292	0.311	0.239	0.316	0.246	0.307	0.251	0.321	0.310	0.339	0.326	0.347
	Avg	**0.198**	**0.242**	**0.198**	0.249	0.233	0.249	0.241	0.275	0.213	0.291	0.222	0.283	0.213	0.295	0.254	0.315	0.265	0.325
1st Count		**27**	**24**	7	5	1	3	0	2	0	0	2	4	0	0	0	0	0	0
Top 2 Count		**34**	**35**	21	20	6	10	8	5	1	0	4	5	1	0	0	0	0	0

* The reproduced baseline results are sourced from Timer_XL. The best results are highlighted in bold red, while the second-best results are highlighted in underlined blue.

temporal dynamics and adaptive variable dependencies, leading to more stable and consistently superior performance in multivariate time series forecasting with long contexts.

4.2. Generalization in short-context forecasting

Although MPGTimer is primarily designed for long-context modeling, it also demonstrates strong generalization capability under the standard short-context forecasting setting with a lookback length of 96. As shown in Table 3, we compare MPGTimer with recent state-of-the-art methods under this widely adopted benchmark. The results indicate that MPGTimer consistently ranks among the top-performing models across most datasets. Even on datasets where MPGTimer did not outperform the average performance of these models, the performance gaps remained marginal, being only on the order of 10^{-3} for the ETT series and Electricity datasets, and approximately 10^{-2} for the Weather and Solar datasets. It should be noted that this setting is intended to ensure a fair comparison with mainstream forecasting models that typically use a lookback length of 96. Since several recent methods are specifically optimized for short-context forecasting, they achieve slightly better performance on a few datasets. Nevertheless, MPGTimer maintains stable and consistent performance in this standard scenario, highlighting its strong generalization across different data distributions rather than being tailored to a specific sequence length.

4.3. Ablation study

4.3.1. Component ablation

To analyze the contribution of key components in MPGTimer, we conduct an ablation study focusing on two modules: the Multi-scale Patch Integrator (MPI) and the Graph-enhanced Interaction Attention (GIA). Specifically, we construct variant models by selectively removing each module and evaluate the resulting performance changes relative to the baseline. *w/o MPI* removes the Multi-scale Patch Integrator and adopts a single-scale patch partitioning scheme for temporal modeling, allowing us to assess the role of multi-scale patches in long-context forecasting. *w/o GIA* replaces the Graph-enhanced Interaction Attention with the TimeAttention mechanism used in Timer_XL, enabling analysis

Table 3
Benchmark results for long-term multivariate forecasting under the standard short-context setting (96-pred-96). This experiment compares MPGTimer with the average performance of recent SOTA models designed for short-context forecasting under standard short lookback length settings, aiming to verify its generalization capability in conventional short-context forecasting scenarios.

Models	MPGTimer (Ours)		Average of baselines		ReCast (2026)		FreEformer (2025)		TQNet (2025)		CSformer (2025)		TimePro (2025)		iTransformer (2024)	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	**0.374**	**0.394**	**0.374**	0.395	0.368	0.387	0.371	0.390	0.371	0.393	0.372	0.394	0.375	0.398	0.386	0.405
ETTh2	0.289	0.341	**0.287**	**0.340**	0.258	0.329	0.286	0.335	0.295	0.343	0.293	0.340	0.293	0.345	0.297	0.349
ETTm1	**0.317**	**0.355**	0.318	0.356	0.308	0.345	0.306	0.340	0.311	0.353	0.324	0.367	0.326	0.364	0.334	0.368
ETTm2	0.175	0.259	**0.173**	**0.257**	0.161	0.243	0.169	0.248	0.173	0.256	0.179	0.269	0.178	0.260	0.180	0.264
Electricity	**0.139**	**0.233**	**0.139**	0.234	0.135	0.234	0.133	0.223	0.134	0.229	0.146	0.242	0.139	0.234	0.148	0.240
Weather	0.174	0.216	**0.160**	**0.201**	0.141	0.181	0.153	0.189	0.157	0.200	0.168	0.217	0.166	0.207	0.174	0.214
Solar-Energy	0.201	0.237	**0.188**	**0.230**	0.181	0.237	0.180	0.191	0.173	0.233	0.197	0.244	0.196	0.237	0.203	0.237

* The reproduced baseline results are sourced from the reported papers. Results are compared based on average performance. The best results are highlighted in bold.

Table 4
Ablation results of MPGTimer. *w/o MPI* denotes the variant without the Multi-scale Patch Integrator, and *w/o GIA* denotes the variant without Graph-enhanced Interaction Attention. *Boost* indicates the percentage performance gain introduced by the corresponding module.

Datasets	ETTh1		ETTh2		ETTm1		ETTm2		Electricity		Weather		Solar-energy	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
w/o MPI	0.406	0.428	0.400	0.432	0.361	0.391	0.264	0.323	0.158	0.251	0.235	0.270	0.230	0.248
MPGTimer	0.393	0.418	0.342	0.392	0.349	0.381	0.257	0.316	0.156	0.250	0.227	0.265	0.198	0.242
Boost	**3.20%**	**2.34%**	**14.50%**	**9.26%**	**3.32%**	**2.56%**	**2.65%**	**2.17%**	**1.27%**	**0.40%**	**3.40%**	**1.85%**	**13.91%**	**2.42%**
w/o GIA	0.401	0.425	0.342	0.393	0.353	0.384	0.270	0.325	0.159	0.252	0.234	0.268	0.240	0.251
MPGTimer	0.393	0.418	0.342	0.392	0.349	0.381	0.257	0.316	0.156	0.250	0.227	0.265	0.198	0.242
Boost	**2.00%**	**1.65%**	–	**0.25%**	**1.13%**	**0.78%**	**4.81%**	**2.77%**	**1.89%**	**0.79%**	**2.99%**	**1.12%**	**17.50%**	**3.59%**

of the contribution of graph-based soft weighting to modeling dynamic variable dependencies and overall predictive performance. In addition, ablation studies on the individual components of GIA are provided in Appendix B.2 to further analyze their contributions to the overall model performance.

Table 4 reports the averaged results of the ablation study, with the complete experimental outcomes provided in Appendix B.1. All evaluations are conducted on the same seven datasets used in the benchmark experiments. Overall, removing any individual module results in performance degradation to varying degrees. When the Multi-scale Patch Integrator (MPI) is removed, the model's ability to capture temporal patterns across different scales deteriorates, with the most significant performance drops observed on the ETTh2 and Solar-Energy datasets. This suggests that MPI plays a critical role in modeling trend variations at multiple temporal resolutions. Without MPI, the model is restricted to single-scale patch representations, leading to a substantial loss in long-context forecasting capability. Replacing the Graph-enhanced Interaction Attention (GIA) with the baseline TimeAttention mechanism similarly degrades performance, particularly in terms of modeling variable dependencies. Since TimeAttention is primarily designed to capture long-range temporal dependencies, its treatment of variable relationships relies on binary masking, which lacks the flexibility to modulate interaction strengths through continuous soft weights. As a result, prediction accuracy decreases markedly on the Solar-Energy dataset, where variable interactions are highly complex. These results further confirm the effectiveness of GIA in capturing variable dependency structures.

The results show that removing any individual module consistently degrades predictive accuracy. In particular, incorporating the Multi-scale Patch Integrator (MPI) yields stable performance gains across all datasets, with average reductions of 6.04% in MSE and 3.00% in MAE. The improvements are especially pronounced on ETTh2, where MSE and MAE decrease by 14.50% and 9.26%, and on Solar-Energy, with reductions of 13.91% and 2.42%. These findings suggest that MPI effectively strengthens multi-scale temporal awareness, enabling the model to capture finer-grained periodic patterns. The Graph-enhanced Interaction Attention (GIA) module also plays a critical role in improving performance. With GIA incorporated, the average MSE and MAE are reduced by 4.33% and 1.56%, respectively. The largest gains are observed on the Solar-Energy dataset, which exhibits high variable interaction complexity, where MSE and MAE decrease by 17.50% and 3.59%. On ETTm2, the corresponding reductions reach 4.81% and 2.77%. These results highlight the effectiveness of GIA in modeling complex multivariate dependencies. Overall, the ablation analysis confirms the effectiveness of MPGTimer. By jointly modeling multi-scale temporal patterns and adaptive variable interactions, the model achieves more stable and accurate forecasting performance in challenging multivariate settings.

4.3.2. Variant analysis

Considering that the optimal scale weighting may vary across different datasets and forecasting horizons, we design an adaptive scale-weighted variant based on the original fusion method to investigate whether patches at different temporal scales contribute differently to the final representation. Specifically, the original MPI completes fusion through a linear mapping after aligning patch embeddings of different scales. In this variant, for each scale's aligned representation, a corresponding learnable scalar parameter is introduced, defined as follows:

$$w_s = \sigma(\alpha_s), \quad \alpha_s \in \mathbb{R} \tag{9}$$

Where α_s represents the learnable parameter corresponding to the s-th scale, and w_s is the scale weight after Sigmoid mapping. In the fusion stage, this weight adaptively adjusts the contribution of information from different temporal scales, thereby enhancing key scale features and suppressing redundant scale interference. Finally, the weighted multi-scale representations are concatenated along the feature dimension and then passed through a linear mapping to generate a unified multi-scale representation.

The results are shown in Table 5. It can be observed that, compared with the original fusion strategy, the adaptive scale weighting variant does not exhibit obvious performance changes on most datasets. This indicates that the unified fusion strategy in the original MPI is already capable of effectively aggregating information from different temporal

Table 5
Variant analysis of MPI scale fusion strategies. *MPI + ASW* denotes a variant that introduces adaptive scale weighting on top of the original MPI fusion method.

Datasets	ETTh1		ETTh2		ETTm1		ETTm2		Electricity		Weather		Solar-energy	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
MPI	0.393	0.418	0.342	0.392	0.349	0.381	0.257	0.316	0.156	0.250	0.227	0.265	0.198	0.242
MPI + ASW	0.392	0.417	0.342	0.392	0.348	0.381	0.260	0.317	0.155	0.248	0.229	0.265	0.205	0.245
Boost	**0.25%**	**0.24%**	–	–	**0.29%**	–	−1.17%	−0.32%	**0.64%**	**0.80%**	−0.88%	–	−3.54%	−1.24%

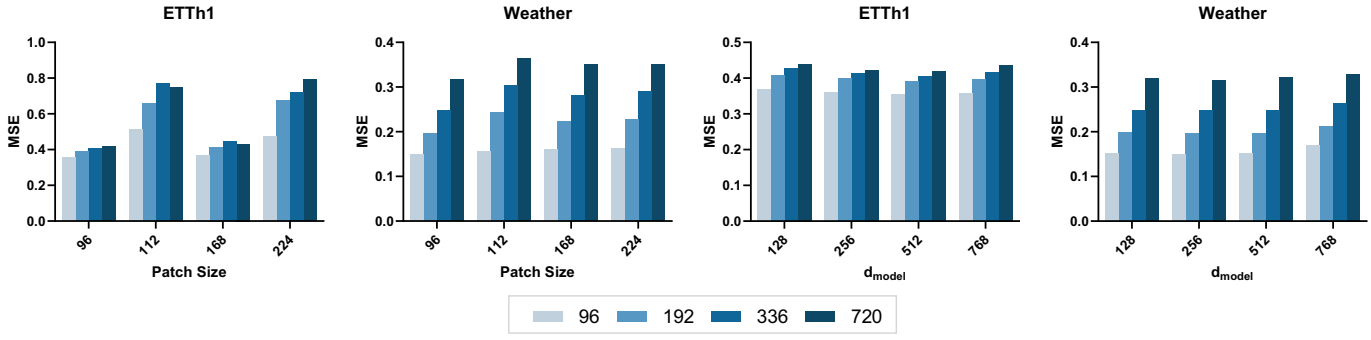


Fig. 4. Hyperparameter sensitivity analysis of the patch size $P = \{96, 112, 168, 224\}$ and the model dimension $d_{\text{model}} = \{128, 256, 512, 768\}$ evaluated on the ETTh1 and weather datasets, with MSE used as the evaluation metric.

scales, and thus the additional scale weighting does not bring significant performance gains. However, on the Solar-Energy dataset, the introduction of adaptive scale weighting leads to a performance degradation. Although both Solar-Energy and Electricity are high-dimensional multivariate datasets, they exhibit notable differences in data distribution. Variables in the Electricity dataset usually exhibit strong synchronization and stable periodicity. Therefore, the importance of different temporal scales can be more easily modeled through adaptive weighting. In contrast, the Solar-Energy dataset is influenced by factors such as weather conditions, geographical locations, and solar irradiance intensity, causing the time series to exhibit stronger fluctuations and non-stationary characteristics. In this case, the multi-scale features themselves already possess strong complementarity. Additional scale recalibration may weaken the collaborative interactions among cross-scale information, thereby leading to performance degradation. Overall, different scale fusion strategies can have a certain impact on model performance. Meanwhile, the effectiveness of adaptive scale weighting varies across different datasets, and its applicable scenarios still require further investigation.

4.4. Hyperparameter analysis

We evaluate the hyperparameter sensitivity of MPGTimer on the ETTh1, Weather, and Electricity datasets, which vary in variable scale from small to large. The hyperparameters considered include the patch size, the model feature dimension d_{model}, and the lookback length, with particular emphasis placed on analyzing the impact of the lookback length on forecasting performance.

In the hyperparameter sensitivity analysis, we follow the same model configuration as in the benchmark experiments and vary the prediction horizon while keeping all other hyperparameters fixed. The corresponding results are presented in Fig. 4. When examining the effect of patch size, the choice of patch size P is subject to two constraints. First, P must evenly divide the lookback length to ensure that the input sequence can be partitioned into an integer number of patches. Second, the kernel sizes at different scales in the Multi-scale Patch Integrator must also be compatible with the lookback length, so as to avoid interpolation errors caused by scale mismatch. Under these constraints, we evaluate four patch sizes, $P = \{96, 112, 168, 224\}$, to assess the impact of patch granularity on forecasting performance. The results indicate that patch size has a substantial influence on prediction accuracy, and the optimal choice depends on both dataset characteristics and the prediction horizon. In most settings, $P = 96$ consistently yields the best performance across different horizons.

To investigate the impact of the model dimension d_{model} on model performance, we vary d_{model} across four settings—128, 256, 512, and 768—and conduct comparative experiments under different forecasting horizons. As shown in Fig. 4, increasing the model dimension from 128 to 256 and 512 consistently improves overall forecasting accuracy, indicating that a moderately larger model dimension enhances the model's representational capacity. However, when d_{model} is further increased to 768, prediction errors exhibit varying degrees of degradation across most forecasting horizons. This suggests that excessively large feature dimensions may introduce redundant representations and increase the difficulty of model optimization. Considering both prediction accuracy and performance stability, intermediate feature dimensions achieve more robust and consistently superior performance across different forecasting scenarios.

In addition, we focus on the impact of the lookback length on model performance. Prior studies have shown that simply increasing the lookback length does not consistently improve forecasting accuracy[23,39]. Consequently, most existing Transformer-based models adopt a relatively short lookback length of 96. Under this setting, we further examine the impact of extending the lookback length on prediction performance. Keeping all other hyperparameter settings consistent with the baseline experiment, we conduct experiments with different lookback lengths across all datasets. For visualization, we select three representative datasets corresponding to scenarios with a small number of variables (ETTh1), a medium number of variables (Weather), and a large number of variables (Electricity). Fig. 5 illustrates that increasing the lookback length yields stable and continuous performance improvements, with optimal performance achieved at a lookback length of approximately 672. Unlike conventional Transformer-based models, which often suffer from performance saturation or degradation when the lookback length becomes large, the proposed model maintains low prediction errors even with substantially longer lookback lengths. Notably, when the lookback length is further extended to ultra-long ranges (≥ 960), the performance gains exhibit diminishing returns. This indicates that the additional

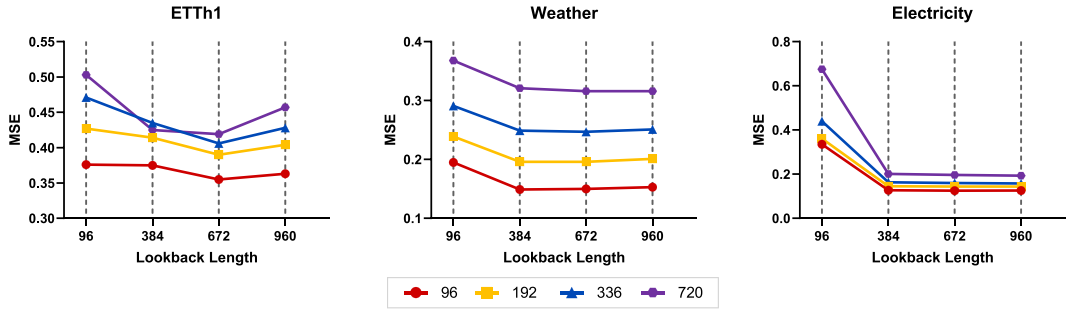


Fig. 5. Comparison of MSE results under different lookback lengths $L = \{96, 384, 672, 960\}$ across multiple forecasting horizons on the ETTh1, weather, and electricity datasets.

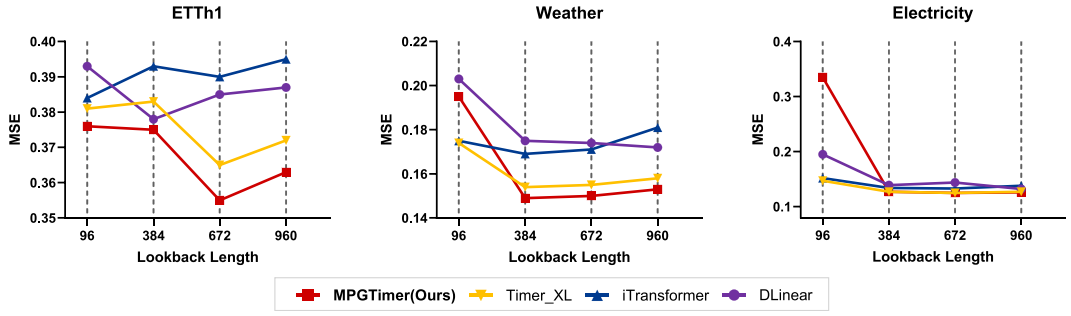


Fig. 6. Comparison of MSE results under different lookback lengths $L = \{96, 384, 672, 960\}$ with the forecasting horizon fixed at 96, evaluating MPGTimer against Timer_XL, DLinear, and iTransformer.

Table 6
Efficiency comparison of different forecasting models on ETTh1, ETTm1, and weather datasets in terms of training time, inference time, parameter size, FLOPs, and peak memory consumption.

Models	Train time (s)			Inference time (s/epoch)			Params (M)			FLOPs (GFLOPs)			Peak memory (GB)		
	ETTh1	ETTm1	Weather	ETTh1	ETTm1	Weather	ETTh1	ETTm1	Weather	ETTh1	ETTm1	Weather	ETTh1	ETTm1	Weather
MPGTimer (Ours)	55.55	250.60	478.96	1.09	3.90	7.75	6.952	6.952	5.713	4.403	4.403	12.211	0.189	0.189	0.191
Timer_XL	44.76	199.42	494.19	0.92	3.45	7.44	2.203	2.203	2.419	3.452	3.452	11.363	0.168	0.168	0.181
Timer	33.15	175.49	219.70	0.71	2.47	4.56	2.202	2.202	2.419	3.452	3.452	11.363	0.062	0.062	0.076
iTransformer	93.51	333.73	471.37	13.93	52.22	66.43	2.498	2.498	2.567	0.055	0.055	0.128	0.039	0.039	0.160
DLinear	58.00	121.88	185.13	1.35	3.62	3.58	0.129	0.129	0.129	0.029	0.029	0.087	0.019	0.018	0.021
Autoformer	137.76	496.02	929.75	6.78	22.02	23.06	5.291	5.291	3.470	68.934	68.934	58.391	0.556	0.513	0.285
Transformer	104.30	350.26	527.41	2.89	9.08	12.82	5.288	5.288	3.463	68.972	68.972	58.426	0.524	0.804	0.750

historical information in excessively long sequences contributes little to prediction, while also increasing the modeling and computational burden. Especially on the Weather and Electricity datasets, although the MSE with lookback lengths of 672 and 960 is quite similar, 960 incurs higher training and inference costs. Therefore, considering both predictive performance and computational efficiency, 672 is a more reasonable choice of lookback length.

Furthermore, we conduct comparative evaluations between MPGTimer and several representative baseline models, including Timer_XL, iTransformer, and DLinear. As illustrated in Fig. 6, MPGTimer achieves superior forecasting performance over these baselines across most prediction horizons in long-context forecasting tasks. Notably, as the lookback length increases, the MSE of MPGTimer exhibits a consistent downward trend. These results indicate that MPGTimer is able to more effectively exploit informative patterns in long sequences, thereby demonstrating enhanced modeling capability and stability in long-context forecasting scenarios.

4.5. Efficiency analysis

To further validate the computational efficiency of MPGTimer, we analyze the training time, inference time, memory consumption, parameter count, and FLOPs on the three datasets where the performance improvements are most significant, namely ETTh1, ETTm1, and Weather. During the experiments, the forecasting horizon is fixed at 96 and the lookback length is set to 672. For fairness, all other hyperparameters are kept consistent with those used in the benchmark experiments. The results are shown in Table 6. To more intuitively demonstrate the differences between efficiency and predictive performance, we further visualize the inference time, memory, and MSE metrics in Fig. 7. As can be observed, MPGTimer exhibits superior computational efficiency and resource utilization compared to other state-of-the-art models, which can be directly attributed to the design of its internal modules. First, during the input encoding stage, MPGTimer adopts a multi-scale patch embedding mechanism. Instead of reconstructing the original sequence through high-dimensional fully connected projections, this module extracts local segments using multiple local window sizes and applies linear mappings to obtain scale-specific representations. Since the number of patches at each scale is approximately L/k, the computational complexity of the linear projections grows linearly with respect to both the number of variables and the number of patches. Moreover, as the number of scales is fixed and remains a small constant, the overall time and space complexity of this module remains linear. In addition,

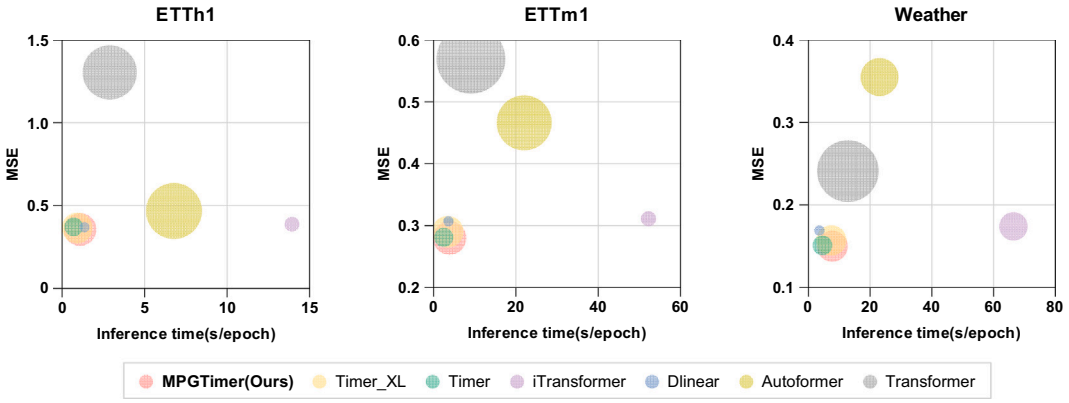


Fig. 7. Comparison of model efficiency on the ETTh1, ETTm1 and weather datasets. Each model utilizes input sequences of length 672 and a forecasting horizon of 96 steps.

when token counts differ across scales, alignment is achieved via average pooling. These operations are performed in a low-dimensional token space and do not introduce additional quadratic interaction costs. Within the backbone network, the proposed TimeAttentionWithGraph module avoids constructing fully connected quadratic correlation matrices along the variable dimension. Instead, it constrains attention weights using an explicit dependency graph. Beyond standard dot-product temporal attention, a graph-attention branch is introduced to model structural correlations among variables through linearly projected node features, which are then fused with temporal attention. This design enables sparse dependency modeling guided by graph structure, such that attention computation is activated only within the dependency subgraph. As a result, attention expansion in high-dimensional multivariate settings is effectively avoided, allowing MPGTimer to balance expressive power and computational tractability.

The experimental results show that, compared with recent state-of-the-art long-context forecasting models, MPGTimer achieves better forecasting performance while maintaining comparable training time, inference time, and GPU memory consumption. Although the model introduces a small amount of additional computational overhead, the overall efficiency remains competitive. Compared with iTransformer, MPGTimer demonstrates superior training and inference efficiency in complex long-context forecasting scenarios. Relative to lightweight models such as DLinear, MPGTimer introduces a more complex structure, resulting in a slight increase in parameter count and inference overhead, but the overall complexity remains within a controllable range. Meanwhile, compared with Transformer and Autoformer, MPGTimer exhibits clearer advantages in both forecasting accuracy and efficiency. Overall, MPGTimer achieves a reasonable trade-off between predictive performance and computational cost, demonstrating its practical potential for long-context time series forecasting tasks.

5. Conclusion and future work

This paper introduces MPGTimer, a novel framework for long-context multivariate time series forecasting. MPGTimer first introduces a Multi-scale Patch Integrator (MPI) to extract local segments at different temporal resolutions, enabling the joint modeling of local patterns and global trends. Building upon this, MPGTimer further designs a Graph-enhanced Interactional Attention (GIA) mechanism that jointly captures temporal and cross-variable dependencies within a unified attention formulation. Specifically, structured masks are employed to constrain interactions among patches, while graph-enhanced attention terms are incorporated to softly model the dependency strengths across multiple variables, thereby improving the representation of complex multivariate dynamics. Extensive experiments on seven representative real-world datasets demonstrate the superiority of MPGTimer, achieving 27 best and 34 top-two results out of 35 MSE metrics, as well as 24 best and 35 top-two results in terms of MAE, with an overall accuracy improvement of 3.13% over state-of-the-art methods. In addition, comprehensive ablation studies, hyperparameter analyses, and efficiency evaluations further validate the critical contributions of the Multi-scale Patch Integrator and the Graph-enhanced Interactional Attention to the model's performance gains.

Future research may proceed along several directions. First, more efficient multi-scale feature construction strategies can be explored, such as dynamically learnable scale partitioning and adaptive patch reorganization, to further enhance the scalability of the model in ultra-long sequence scenarios. Second, building upon the existing Graph-enhanced Interaction Attention mechanism, natural language processing models can be incorporated to strengthen causal reasoning and interpretability. Specifically, large language models can be leveraged to parse textual event descriptions, operational logs, or domain knowledge, and to map high-level semantic causal information into structured constraints or prior biases. These semantic priors can then be integrated with the causal structure matrices and graph-enhanced attention mechanisms in MPGTimer, thereby introducing causally informed reasoning capabilities with semantic interpretability. Finally, deeper integration with multimodal fusion mechanisms could be investigated to extend the applicability of MPGTimer to more complex cross-domain and multimodal time series forecasting tasks. Through these research directions, future work aims to further improve the accuracy, stability, and generalization ability of MPGTimer, contributing to the continued advancement of multivariate time series forecasting methods.

CRediT authorship contribution statement

Xinyi Duan: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Xing Wu:** Writing – review & editing, Supervision, Resources. **Ying Zhou:** Writing – review & editing, Investigation. **Guangpu Jiang:** Supervision, Project administration. **Kang Tu:** Supervision, Project administration. **Zhiye Xu:** Supervision, Resources. **Pengcheng Wu:** Supervision, Resources. **Xixi Liu:** Supervision, Resources. **Jianbiao Dai:** Validation, Supervision.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the author(s) used a generative artificial intelligence tool ChatGPT (OpenAI, GPT-5.2) for language editing and grammatical checking. All content was reviewed and approved by the author(s), who take full responsibility for the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported by the Special Project for High-Quality Development (DZBM2024-31-0022).

Appendix A. Experimental details

A.1. Datasets

We select seven benchmark datasets widely used in multivariate time series forecasting (MTSF). Detailed descriptions of these datasets are provided in Table A.1.

ETT[1]: The Electrical Transformer Temperature (ETT) dataset is collected from power transformer monitoring systems in two regions of China. Its subsets are defined by sampling frequency: ETTh1 and ETTh2 are sampled at an hourly resolution, whereas ETTm1 and ETTm2 are sampled at 15-minute intervals.

Electricity[2]: This dataset contains electricity consumption records from 321 customers in France collected between 2012 and 2014. Electricity usage is measured in kilowatt-hours (kWh) and sampled at an hourly resolution.

Weather[3]: The Weather dataset consists of meteorological observations collected throughout 2020 from a weather station operated by the Max Planck Institute for Biogeochemistry in Germany. It includes 21 meteorological variables recorded at a 10-minute resolution.

Solar-Energy[4]: This dataset contains solar power generation measurements collected from 137 photovoltaic power stations in Alabama in 2006, sampled at 10-minute intervals.

Table A.1
Dataset details. Dim. denotes the number of variables. Training setting indicates the lookback length and forecasting horizon. Dataset length reports the numbers of samples in the training, validation, and test sets.

Dataset	Timesteps	Dim.	Training setting	Dataset length	Freq
ETTh1	17,420	7	672→{96,192,336,720}	(8545, 2881, 2881)	Hourly
ETTh2	17,420	7	672→{96,192,336,720}	(8545, 2881, 2881)	Hourly
ETTm1	69,680	7	672→{96,192,336,720}	(34465, 11,521, 11521)	15 mins
ETTm2	69,680	7	672→{96,192,336,720}	(34465, 11,521, 11521)	15 mins
Electricity	26,304	321	672→{96,192,336,720}	(18317, 2633, 5261)	Hourly
Weather	52,696	21	672→{96,192,336,720}	(36792, 5271, 10540)	10 mins
Solar-Energy	52,560	137	672→{96,192,336,720}	(36601, 5161, 10417)	10 mins

A.2. Baseline settings

Under the long-context setting, we compare our model with eight advanced baselines that are widely recognized in multivariate time series forecasting.

- Timer_XL: A Transformer-based time series model optimized for long-sequence forecasting. It improves the modeling of ultra-long-term dependencies by introducing efficient temporal encodings and long-range attention mechanisms. The code is available in this repository: https://github.com/thuml/Timer-XL.
- Timer: A Transformer-based architecture designed for long-sequence forecasting. It incorporates temporal convolutions and multi-scale attention mechanisms to capture patterns at different temporal resolutions. The code is available in this repository: https://github.com/thuml/Timer.
- UniTST: A unified Transformer-based framework for time series forecasting. It integrates multi-task learning with multi-scale attention to jointly handle diverse tasks, such as regression and classification.
- iTransformer: A Transformer-based architecture with inverted dimensions. It employs variables as tokens, thereby circumventing the issue of historical information decay caused by excessive temporal spans. The code is available in this repository: https://github.com/thuml/iTransformer.
- DLinear: A linear forecasting framework based on time series decomposition. It decomposes the input sequence into trend and residual components, which are modeled using separate linear layers and then aggregated to produce the final forecast. The code is available in this repository: https://github.com/cure-lab/LTSF-Linear.
- TimesNet: A time series forecasting framework designed for multi-task learning. It captures both intra-period and inter-period variations by transforming 1D time series into a set of 2D representations across multiple temporal periods. The code is available in this repository: https://github.com/thuml/TimesNet.
- Stationary: A Transformer-based framework designed for non-stationary time series forecasting. It enhances predictive performance by integrating series stationarization with a de-stationary attention mechanism, enabling the model to adapt to distribution shifts over time. The code is available in this repository: https://github.com/thuml/Nonstationar_Transformers.
- Autoformer: A long-sequence time series forecasting model based on autocorrelation mechanisms and an embedded sequence decomposition module. It captures long-range temporal dependencies at the sub-sequence level, enabling efficient modeling of complex temporal patterns. The code is available in this repository: https://github.com/thuml/Autoformer.

Under the standard short-context setting, we compare our model with six state-of-the-art models in multivariate time series forecasting in recent years. The details are as follows:

- ReCast: A lightweight time series forecasting model based on a codebook mechanism. It achieves efficient and robust forecasting through a residual dual-branch structure and a reliability-aware update mechanism.
- FreEformer: A multivariate time series forecasting model that combines frequency-domain characteristics with an enhanced attention mechanism. It achieves more robust temporal modeling through an additional learnable matrix and an L1 normalization strategy. The code is available in this repository: https://github.com/jackyue1994/FreEformer.
- TQNet: A time series forecasting model based on periodic learnable queries (TQ). It models global variable dependencies through TQ and achieves efficient complex temporal forecasting by integrating single-layer attention with lightweight MLP structures. The code is available in this repository: https://github.com/ACAT-SCUT/TQNet.
- CSformer: A multivariate time series forecasting model based on a two-stage self-attention mechanism. It achieves robust complex temporal forecasting by combining adapter techniques with channel-independent modeling followed by channel-mixing interactions.
- TimePro: A time series forecasting model based on variate-aware and time-aware hyper-state modeling. It efficiently captures long-term temporal dependencies through the construction of hyper-states. The code is available in this repository: https://github.com/xwmaxwma/TimePro.

[1] https://github.com/zhouhaoyi/ETDataset.
[2] https://archive.ics.uci.edu/ml/datasets/ElectricityLoadDiagrams20112014.
[3] https://www.bgc-jena.mpg.de/wetter/.
[4] http://www.nrel.gov/grid/solar-power-data.html.

Table A.2
Hyperparameter configuration for MPGTimer under the long-context scenario. L denotes the number of layers in the transformer, D represents the token embedding dimension, D_{ff} is the hidden dimension of the feedforward sublayer, H indicates the number of heads in the multi-head self-attention, P signifies the patch size, and LR denotes the initial learning rate.

Dataset	Configuration					Training process			
	L	D	D_{ff}	H	P	LR	Dropout	Batch size	Epochs
ETTh1	1	512	1024	8	96	0.0001	0.2	16	10
ETTh2	1	512	1024	8	96	0.0001	0.9	8	10
ETTm1	1	512	1024	8	96	0.0001	0.4	16	10
ETTm2	1	512	1024	8	96	0.0001	0.3	32	10
Electricity	4	512	1024	8	96	0.001	0.1	4	10
Weather	3	256	1024	8	96	0.0005	0.2	16	10
Solar-Energy	6	512	2048	8	96	0.0001	0.1	4	10

Table A.3
Hyperparameter configuration for MPGTimer under the short-context scenario. L denotes the number of layers in the transformer, D represents the token embedding dimension, D_{ff} is the hidden dimension of the feedforward sublayer, H indicates the number of heads in the multi-head self-attention, P signifies the patch size, and LR denotes the initial learning rate.

Dataset	Configuration					Training process			
	L	D	D_{ff}	H	P	LR	Dropout	Batch size	Epochs
ETTh1	1	512	1024	8	96	0.0001	0.2	8	10
ETTh2	1	512	1024	8	96	0.0001	0.2	8	10
ETTm1	1	512	1024	8	96	0.0001	0.4	8	10
ETTm2	1	512	1024	8	96	0.0001	0.3	8	10
Electricity	6	512	1024	8	96	0.0005	0.1	4	10
Weather	1	256	1024	8	96	0.0005	0.2	32	10
Solar-Energy	6	512	2048	8	96	0.0001	0.2	4	10

- iTransformer: A Transformer-based architecture with inverted dimensions. It employs variables as tokens, thereby circumventing the issue of historical information decay caused by excessive temporal spans. The code is available in this repository: https://github.com/thuml/iTransformer.

A.3. Hyperparameter details

Tables A.2 and A.3 summarize the hyperparameter settings under long-context and short-context scenarios, respectively. The tables report the tunable hyperparameters involved in the training process, such as the learning rate and batch size. These hyperparameters are tuned according to different experimental scenarios and datasets to ensure stable training and optimal predictive performance.

Appendix B. Supplementary results

B.1. Comprehensive ablation results

Tables B.1 and B.2 present the comprehensive results of the MPGTimer ablation study. All models are trained with a lookback length of 672 and a forecasting horizon of 96 under a unified experimental setting. During inference, a rolling forecasting strategy is consistently adopted to generate final predictions over different forecasting horizons of $\{96, 192, 336, 720\}$. By systematically replacing and removing key modules, we evaluate the effectiveness and enhancement impact of each component within the model.

B.2. Individual component analysis

To provide a more detailed understanding of the contributions of individual components within the GIA module, we conduct a fine-grained ablation study on different structures involved in the attention computation. Specifically, the original self-attention mechanism computes attention scores solely based on content similarity through $\mathbf{QK}^\top$. Building upon this formulation, we further incorporate the structural mask matrix $\mathbf{M}$ and the graph relation enhancement term $\mathbf{G}$ to progressively improve the modeling of local structural dependencies and cross-variable correlations. Accordingly, we evaluate the following three configurations separately:

1. Using only the original attention term $\mathbf{QK}^\top$;
2. Using $\mathbf{QK}^\top + \mathbf{M}$;
3. Using the complete formulation $\mathbf{QK}^\top + \mathbf{M} + \mathbf{G}$.

Here, $\mathbf{M}$ imposes structural constraints on attention connections, enabling the model to focus more effectively on token interactions with meaningful temporal dependencies. Meanwhile, $\mathbf{G}$ further incorporates graph relational information to enhance the modeling of global correlations across variables. By progressively introducing $\mathbf{M}$ and $\mathbf{G}$, the individual contributions of different components to forecasting performance can be more clearly evaluated. We conduct experiments on the same datasets as those used in the benchmark experiments, with all experimental settings and parameter configurations kept consistent. The results are presented in Table B.3.

Table B.1
Comprehensive ablation results of MPGTimer on the multi-scale patch integrator (MPI).

Horizon		96		192		336		720	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	w/o MPI	0.364	0.397	0.404	0.423	0.423	0.437	0.433	0.455
	MPGTimer	0.355	0.389	0.390	0.412	0.406	0.424	0.419	0.447
	Boost	**2.47%**	**2.02%**	**3.47%**	**2.60%**	**4.02%**	**2.97%**	**3.23%**	**1.76%**
ETTh2	w/o MPI	0.297	0.361	0.379	0.417	0.436	0.457	0.487	0.493
	MPGTimer	0.274	0.341	0.333	0.382	0.360	0.406	0.401	0.440
	Boost	**7.74%**	**5.54%**	**12.14%**	**8.39%**	**17.43%**	**11.16%**	**17.66%**	**10.75%**
ETTm1	w/o MPI	0.291	0.349	0.339	0.377	0.377	0.401	0.438	0.437
	MPGTimer	0.280	0.339	0.327	0.368	0.364	0.390	0.424	0.425
	Boost	**3.78%**	**2.87%**	**3.54%**	**2.39%**	**3.45%**	**2.74%**	**3.20%**	**2.75%**
ETTm2	w/o MPI	0.173	0.264	0.230	0.302	0.283	0.337	0.369	0.390
	MPGTimer	0.169	0.256	0.224	0.295	0.274	0.329	0.361	0.385
	Boost	**2.31%**	**3.03%**	**2.61%**	**2.32%**	**3.18%**	**2.37%**	**2.17%**	**1.28%**
Electricity	w/o MPI	0.126	0.221	0.146	0.240	0.163	0.257	0.196	0.286
	MPGTimer	0.125	0.220	0.144	0.238	0.160	0.254	0.196	0.286
	Boost	**0.79%**	**0.45%**	**1.37%**	**0.83%**	**1.84%**	**1.17%**	–	–
Weather	w/o MPI	0.153	0.203	0.200	0.247	0.254	0.289	0.331	0.342
	MPGTimer	0.150	0.200	0.196	0.243	0.247	0.283	0.316	0.332
	Boost	**1.96%**	**1.48%**	**2.00%**	**1.62%**	**2.76%**	**2.08%**	**4.53%**	**2.92%**
Solar-Energy	w/o MPI	0.173	0.210	0.205	0.233	0.237	0.255	0.306	0.293
	MPGTimer	0.163	0.213	0.189	0.233	0.206	0.249	0.235	0.274
	Boost	**5.78%**	−1.43%	**7.80%**	–	**13.08%**	**2.35%**	**23.20%**	**6.48%**

Table B.2
Comprehensive ablation results of MPGTimer on the graph-enhanced interaction attention (GIA).

Horizon		96		192		336		720	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	w/o GIA	0.360	0.393	0.398	0.418	0.417	0.432	0.427	0.455
	MPGTimer	0.355	0.389	0.390	0.412	0.406	0.424	0.419	0.447
	Boost	**1.39%**	**1.02%**	**2.01%**	**1.44%**	**2.64%**	**1.85%**	**1.87%**	**1.76%**
ETTh2	w/o GIA	0.274	0.341	0.332	0.382	0.359	0.406	0.401	0.441
	MPGTimer	0.274	0.341	0.333	0.382	0.360	0.406	0.401	0.440
	Boost	–	–	−0.3%	–	−0.28%	–	–	**0.23%**
ETTm1	w/o GIA	0.284	0.341	0.331	0.371	0.368	0.393	0.428	0.430
	MPGTimer	0.280	0.339	0.327	0.368	0.364	0.390	0.424	0.425
	Boost	**1.41%**	**0.59%**	**1.21%**	**0.81%**	**1.09%**	**0.76%**	**0.93%**	**1.16%**
ETTm2	w/o GIA	0.179	0.264	0.237	0.304	0.292	0.340	0.373	0.393
	MPGTimer	0.169	0.256	0.224	0.295	0.274	0.329	0.361	0.385
	Boost	**5.59%**	**3.03%**	**5.49%**	**2.96%**	**6.16%**	**3.24%**	**3.22%**	**2.04%**
Electricity	w/o GIA	0.127	0.221	0.146	0.239	0.162	0.257	0.200	0.289
	MPGTimer	0.125	0.220	0.144	0.238	0.160	0.254	0.196	0.286
	Boost	**1.57%**	**0.45%**	**1.37%**	**0.42%**	**1.23%**	**1.17%**	**2.00%**	**1.04%**
Weather	w/o GIA	0.151	0.200	0.200	0.245	0.254	0.287	0.329	0.339
	MPGTimer	0.150	0.200	0.196	0.243	0.247	0.283	0.316	0.332
	Boost	**0.66%**	-	**2.00%**	**0.82%**	**2.76%**	**1.39%**	**3.95%**	**2.06%**
Solar-Energy	w/o GIA	0.184	0.214	0.217	0.237	0.242	0.256	0.317	0.296
	MPGTimer	0.163	0.213	0.189	0.233	0.206	0.249	0.235	0.274
	Boost	**11.41%**	**0.47%**	**12.90%**	**1.69%**	**14.88%**	**2.73%**	**25.87%**	**7.43%**

The experimental results demonstrate that the forecasting performance consistently improves with the progressive incorporation of structural information. Compared with the original attention mechanism based solely on content similarity $\mathbf{QK}^\top$, introducing the structural mask matrix $\mathbf{M}$ enables more stable dependency modeling, reducing the MSE and MAE metrics by 2.93% and 2.03%, respectively. Building upon this, the further integration of the graph relation enhancement term $\mathbf{G}$ improves the model's ability to capture complex inter-variable correlation patterns, achieving an additional 4.04% reduction in MSE compared with the $\mathbf{QK}^\top + \mathbf{M}$ structure, thereby yielding the best overall forecasting performance. Moreover, the performance gains become increasingly pronounced as the prediction horizon grows, indicating that the proposed GIA module is more effective in alleviating long-term dependency modeling difficulties and error accumulation in long-horizon forecasting scenarios. Overall, the complete $\mathbf{QK}^\top + \mathbf{M} + \mathbf{G}$ structure achieves superior forecasting performance, validating the effectiveness and complementarity of the individual components within the GIA module.

Table B.3
Comprehensive ablation results of different components in the GIA module.

Horizon		96		192		336		720	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	$QK^\top$	0.360	0.394	0.399	0.420	0.422	0.436	0.447	0.465
	$QK^\top + M$	0.360	0.393	0.398	0.418	0.417	0.432	0.427	0.455
	$QK^\top + M + G$	**0.355**	**0.389**	**0.390**	**0.412**	**0.406**	**0.424**	**0.419**	**0.447**
ETTh2	$QK^\top$	0.275	0.342	0.333	**0.382**	**0.359**	**0.406**	**0.400**	**0.440**
	$QK^\top + M$	**0.274**	**0.341**	**0.332**	**0.382**	**0.359**	**0.406**	**0.401**	**0.441**
	$QK^\top + M + G$	**0.274**	**0.341**	0.333	**0.382**	0.360	**0.406**	0.401	**0.440**
ETTm1	$QK^\top$	0.283	0.342	0.335	0.374	0.379	0.402	0.454	0.445
	$QK^\top + M$	0.284	0.341	0.331	0.371	0.368	0.393	0.428	0.430
	$QK^\top + M + G$	**0.280**	**0.339**	**0.327**	**0.368**	**0.364**	**0.390**	**0.424**	**0.425**
ETTm2	$QK^\top$	0.178	0.267	0.240	0.312	0.298	0.350	0.392	0.408
	$QK^\top + M$	0.179	0.264	0.237	0.304	0.292	0.340	0.373	0.393
	$QK^\top + M + G$	**0.169**	**0.256**	**0.224**	**0.295**	**0.274**	**0.329**	**0.361**	**0.385**
Electricity	$QK^\top$	0.142	0.235	0.160	0.253	0.180	0.271	0.219	0.305
	$QK^\top + M$	0.127	0.221	0.146	0.239	0.162	0.257	0.200	0.289
	$QK^\top + M + G$	**0.125**	**0.220**	**0.144**	**0.238**	**0.160**	**0.254**	**0.196**	**0.286**
Weather	$QK^\top$	0.167	0.215	0.214	0.259	0.265	0.297	0.336	0.346
	$QK^\top + M$	0.151	**0.200**	0.200	0.245	0.254	0.287	0.329	0.339
	$QK^\top + M + G$	**0.150**	**0.200**	**0.196**	**0.243**	**0.247**	**0.283**	**0.316**	**0.332**
Solar-Energy	$QK^\top$	0.210	0.223	0.234	0.242	0.248	0.256	0.256	**0.270**
	$QK^\top + M$	0.184	0.214	0.217	0.237	0.242	0.256	0.317	0.296
	$QK^\top + M + G$	**0.163**	**0.213**	**0.189**	**0.233**	**0.206**	**0.249**	**0.235**	0.274

* The best results are highlighted in bold red, while the second-best results are highlighted in underlined blue.

Appendix C. Showcases

We visualize the forecasting results on datasets with different variable scales, including the ETTh1 dataset with 7 variables and the Electricity dataset with 321 variables, and compare them against several representative time series forecasting models. In each figure, the blue curve denotes the ground-truth values of the target series, while the orange curve represents the corresponding predictions. As shown in the visualizations, MPGTimer exhibits superior forecasting accuracy. Specifically, as illustrated in Fig. C.1, on the ETTh1 dataset with a relatively small number of variables, MPGTimer closely follows the ground-truth sequence in terms of overall trend. It also accurately captures the locations of extreme values. In contrast, Timer, iTransformer, and DLinear show noticeable deviations in modeling fluctuations. While Autoformer and Transformer can reflect overall trend changes, their prediction curves exhibit limited fitting accuracy, making it difficult to accurately match the magnitude of actual fluctuations. On the Electricity dataset, which contains 321 variables, MPGTimer also demonstrates robust forecasting performance. As observed in Fig. C.2, even in settings with a large number of variables, MPGTimer is able to accurately track the dominant trends and fluctuations of the target series. In contrast,

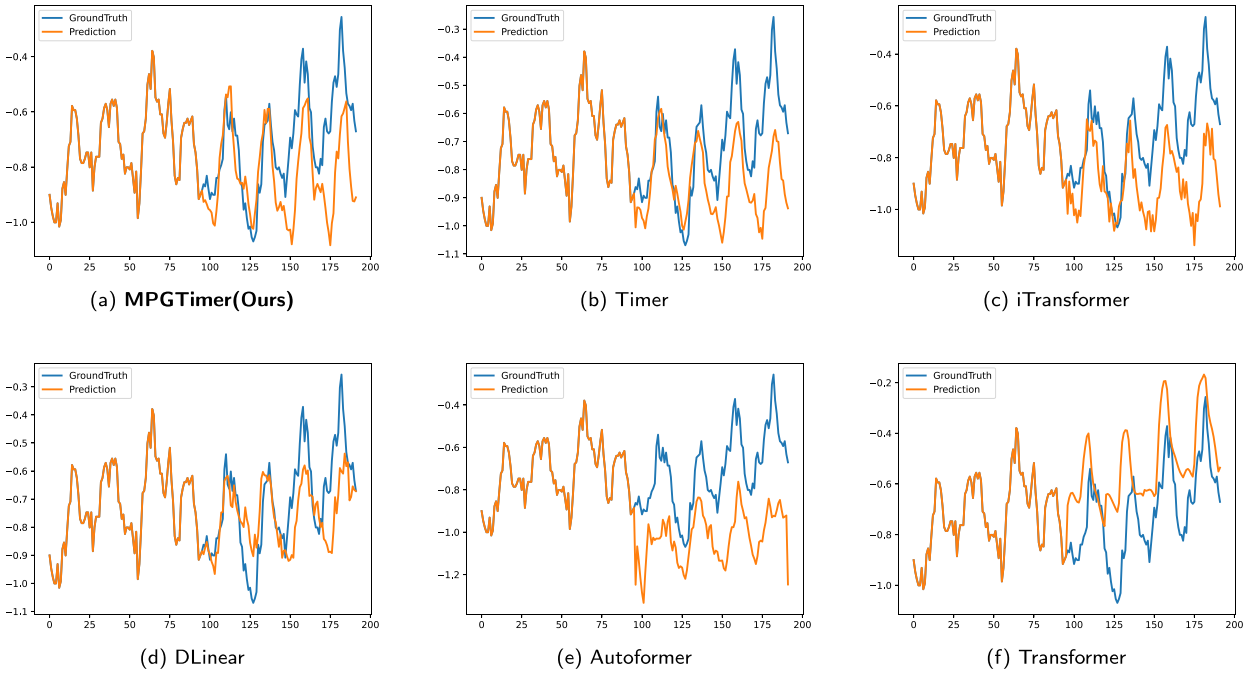


Fig. C.1. Visualization results on the ETTh1 dataset (7 variables) under the input-672-pred-96 setting.

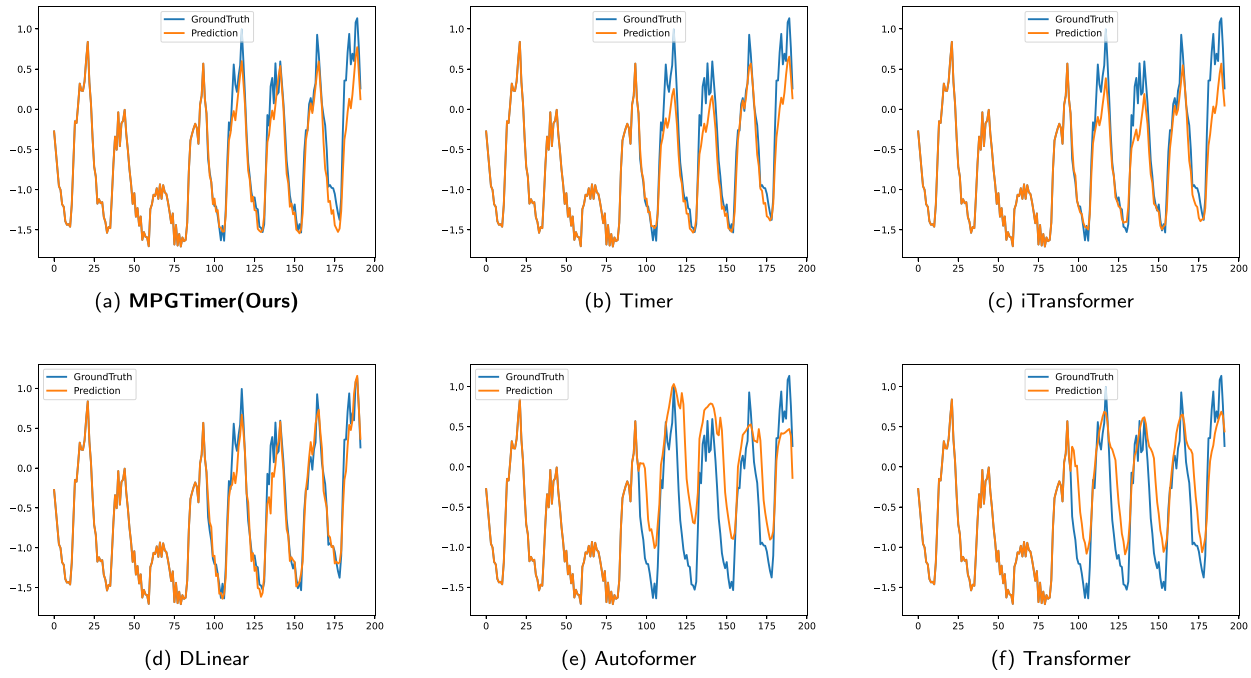


Fig. C.2. Visualization results on the electricity dataset (321 variables) under the input-672-pred-96 setting.

the predictions produced by other models exhibit varying degrees of deviation when modeling sequence fluctuations. The visualization results show that MPGTimer maintains strong forecasting performance across different variable scales, indicating its potential for real-world applications.

Data availability

The seven publicly available datasets used in this study are available from a public repository (https://huggingface.co/datasets/thuml/Time-Series-Library).

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